

Results

01 • EXPLORATORY FACTOR ANALYSIS (EFA)

discover underlying factors

Table 1. Suitability

KMO	Bartlett's χ^2	df	p
.75	904.1	36	<.001

Table 2. Variance explained

Factor	Eigenvalue	% variance	Cumulative %
F1	2.24	24.9	24.9
F2	1.34	14.9	39.8
F3	1.28	14.2	54.0

Table 3. Rotated factor loadings

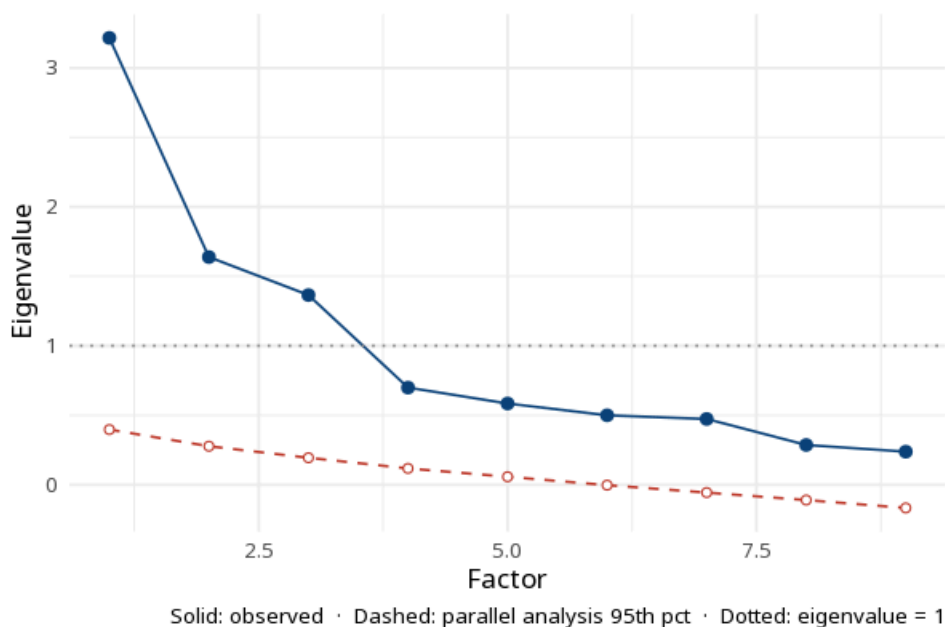
Item	F1	F2	F3	Communality
x1		.59		.48
x2		.51		.26
x3		.69		.45
x4	.85			.73
x5	.89			.75
x6	.81			.69
x7			.73	.51
x8			.69	.52
x9		.38	.46	.46

Table 4. Interfactor correlations (Φ)

	F1	F2	F3
F1	1.00		
F2	0.32	1.00	
F3	0.21	0.26	1.00

Loading columns expand to the number of retained factors; loadings < .32| suppressed (Tabachnick & Fidell). Φ (interfactor correlations) shown for oblique rotation (oblimin) only — omitted for varimax (orthogonal). $KMO \geq .60$ acceptable, $\geq .70$ preferred (Kaiser & Rice, 1974). Bartlett $p < .001$ required. Parallel analysis (Horn, 1965) is the preferred retention rule; Kaiser eigenvalue > 1 is offered but systematically over-extracts (Zwick & Velicer, 1986). Factor order: descending SS-loadings.

Figure. Factor retention



How to read this test

Check KMO ($> .60$ acceptable, $> .70$ preferred; Kaiser & Rice, 1974) and a significant Bartlett's test ($p < .001$) for suitability. Parallel analysis (Horn, 1965) is the recommended retention rule — it corrects the upward bias of the Kaiser eigenvalue > 1 rule (Zwick & Velicer, 1986). Table 3 shows the rotated pattern matrix: each loading is the unique contribution of the factor to that item. Loadings $< .32$ ($\sim 10\%$ variance) are suppressed; a loading $\geq .32$ on two or more factors flags a cross-loader (Tabachnick & Fidell). For oblique rotation (oblimin), the Φ matrix shows interfactor correlations — if all $|\Phi| < .32$, orthogonal rotation (varimax) may be preferred. EFA is exploratory — it discovers structure; to confirm a proposed factor structure run a CFA, ideally on a separate sample (Watkins, 2018).

APA template: EFA (KMO = .75, Bartlett's $\chi^2(36) = 904.1$, $p < .001$) with parallel analysis retained 3 factors explaining 54.0% of variance (oblimin rotation).

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